JSON with Bytes as Base 64

If your app needs to receive and send JSON data, but you need to include binary data in it, you can encode it as base 64.

Base 64 vs Files

Consider first if you can use Request Files for uploading binary data and Custom Response - FileResponse for sending binary data, instead of encoding it in JSON.

JSON can only contain UTF-8 encoded strings, so it can't contain raw bytes.

Base 64 can encode binary data in strings, but to do it, it needs to use more characters than the original binary data, so it would normally be less efficient than regular files.

Use base 64 only if you definitely need to include binary data in JSON, and you can't use files for that.

Pydantic `bytes`

You can declare a Pydantic model with `bytes` fields, and then use `val_json_bytes` in the model config to tell it to use base 64 to *validate* input JSON data, as part of that validation it will decode the base 64 string into bytes.

Python 3.10+

```python
from fastapi import FastAPI
from pydantic import BaseModel


class DataInput(BaseModel):
    description: str
    data: bytes

    model_config = {"val_json_bytes": "base64"}

# Code here omitted 👈

app = FastAPI()


@app.post("/data")
def post_data(body: DataInput):
    content = body.data.decode("utf-8")
    return {"description": body.description, "content": content}

# Code below omitted 👇
```

Ask AI

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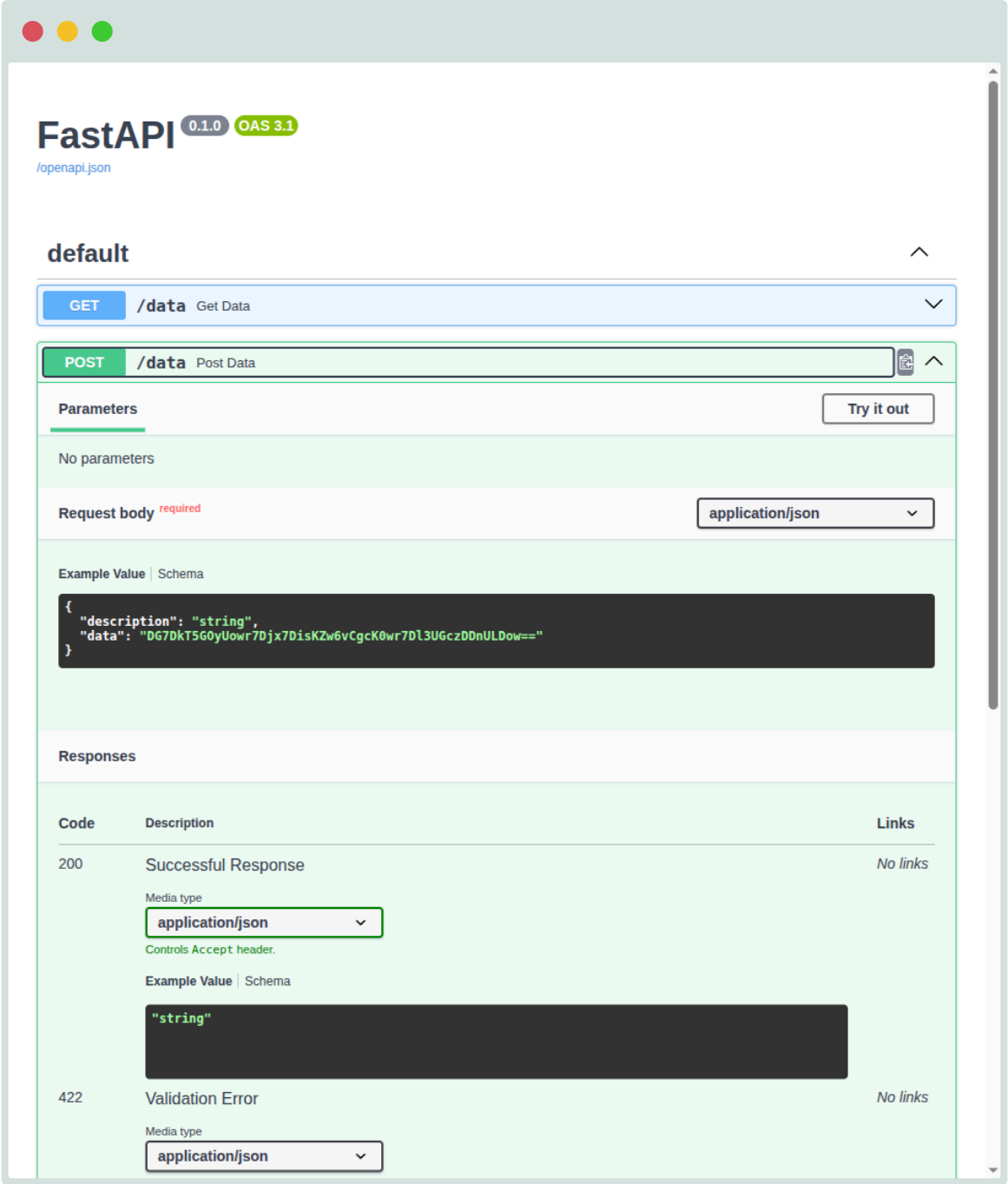
Python 3.10+

```python
from fastapi import FastAPI
from pydantic import BaseModel


class DataInput(BaseModel):
    description: str
    data: bytes

    model_config = {"val_json_bytes": "base64"}


class DataOutput(BaseModel):
    description: str
    data: bytes

    model_config = {"ser_json_bytes": "base64"}


class DataInputOutput(BaseModel):
    description: str
    data: bytes

    model_config = {
        "val_json_bytes": "base64",
        "ser_json_bytes": "base64",
    }


app = FastAPI()


@app.post("/data")
def post_data(body: DataInput):
    content = body.data.decode("utf-8")
    return {"description": body.description, "content": content}


@app.get("/data")
def get_data() -> DataOutput:
    data = "hello".encode("utf-8")
    return DataOutput(description="A plumbus", data=data)


@app.post("/data-in-out")
def post_data_in_out(body: DataInputOutput) -> DataInputOutput:
    return body
```

If you check the `/docs`, they will show that the field `data` expects base 64 encoded bytes:



You could send a request like:

```
{
    "description": "Some data",
    "data": "aGVsbG8="
}
```

Tip

`aGVsbG8=` is the base 64 encoding of `hello`.

And then Pydantic will decode the base 64 string and give you the original bytes in the `data` field of the model.

You will receive a response like:

```
{
  "description": "Some data",
  "content": "hello"
}
```

Pydantic `bytes` for Output Data

You can also use `bytes` fields with `ser_json_bytes` in the model config for output data, and Pydantic will *serialize* the bytes as base `64`. when generating the JSON response.

Python 3.10+

```python
from fastapi import FastAPI
from pydantic import BaseModel

# Code here omitted 👆

class DataOutput(BaseModel):
    description: str
    data: bytes

    model_config = {"ser_json_bytes": "base64"}

# Code here omitted 👆

app = FastAPI()

# Code here omitted 👆

@app.get("/data")
def get_data() -> DataOutput:
    data = "hello".encode("utf-8")
    return DataOutput(description="A plumbus", data=data)

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👀 **Full file preview**

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    return DataOutput(description="A plumbus", data=data)


@app.post("/data-in-out")
def post_data_in_out(body: DataInputOutput) -> DataInputOutput:
    return body
```

Pydantic `bytes` for Input and Output Data

And of course, you can use the same model configured to use base `64` to handle both input (*validate*) with `val_json_bytes` and output (*serialize*) with `ser_json_bytes` when receiving and sending JSON data.

Python 3.10+

```python
from fastapi import FastAPI
from pydantic import BaseModel

# Code here omitted 👈

class DataInputOutput(BaseModel):
    description: str
    data: bytes

    model_config = {
```

```
        "val_json_bytes": "base64",
        "ser_json_bytes": "base64",
    }

# Code here omitted 👈

app = FastAPI()

# Code here omitted 👈

@app.post("/data-in-out")
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